Nanoparticle-Mediated VCP Inhibition Confers a Mutation-Independent Neuroprotection across Five PDE6-Associated Autosomal Recessive Models of Retinitis Pigmentosa

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Background:
Retinitis pigmentosa is a genetically heterogeneous inherited retinal degeneration characterized by progressive photoreceptor loss. Although caused by diverse mutations, disease progression converges on disrupted proteostasis and cellular stress pathways. The ATP-driven chaperone valosin-containing protein (VCP), a central regulator of protein quality control, represents a promising therapeutic target due to its involvement in these convergent pathways. We previously demonstrated that pharmacological VCP inhibition slows degeneration in rhodopsin-associated autosomal dominant models. While preliminary evidence supported efficacy in Pde6b-associated degeneration, its impact on autosomal recessive Pde6a mutations remained unclear.

Here, we employed an organotypic retinal explant platform to evaluate the gene-agnostic potential of sustained VCP inhibition across five distinct PDE6-associated recessive models.

Methods:
Organotypic retinal explants were prepared from five mouse models: $PDE6b^{rd10}$, $PDE6a^{D670G}$, $PDE6a^{R562W}$, $PDE6a^{V685M}$, and $PDE6a^{R562W/V685M}$. Explants were cultured *in vitro* in a defined serum-free medium and treated on day in vitro 0 with escalating concentrations of $mPEG_{5kDa}$-cholane-encapsulated ML240. Empty nanoparticles served as vehicle controls. Retinal pigment epithelium (RPE) was preserved during explant preparation to maintain outer retinal structural integrity and physiological relevance. Degeneration stage-matched tissues were processed for immunostaining. Neuroprotection was quantified by measuring photoreceptor row number in the outer nuclear layer (ONL). The TUNEL assay assessed cell death.

Results:
Treatment with $mPEG_{5kDa}$-cholane-encapsulated ML240 showed no detectable toxicity across all five models. Notably, a statistically significant decrease in TUNEL-positive cells was detected in the $PDE6a^{R562W}$ model. The overall response pattern supports mutation-independent attenuation of degeneration in PDE6-associated recessive contexts.

Conclusions:
These findings demonstrate that organotypic retinal explants provide a robust comparative platform to evaluate therapeutic responses across genetically distinct models. VCP inhibition confers neuroprotection in multiple PDE6-associated autosomal recessive genotypes, supporting its potential as a gene-agnostic therapeutic strategy in inherited retinal degeneration.

Importantly, this culture system enables controlled evaluation of sustained-release nanoparticle delivery systems, extending its applicability beyond soluble compounds and strengthening its value for translational preclinical assessment.